

Synergetic effect of epoxy resin and maleic anhydride grafted polypropylene on improving mechanical properties of polypropylene/short carbon fiber composites



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ABSTRACT

The influence of epoxy resin (EP) and maleic anhydride grafted polypropylene (PPMA) on mechanical properties of polypropylene/short carbon fiber (PP/SCF) composites was studied. The combination between EP and PPMA showed a synergistic effect on improving the mechanical properties of PP/SCF composites, including tension, flexural and notched impact properties. The synergistic mechanism was studied by means of SEM observation, FTIR analysis, extraction test and rheological measurement. The results showed that the synergistic mechanism was attributed to the improved network structure of SCF in PP matrix assisted by EP and PPMA. EP adsorbed onto the surface of SCF reacted with PPMA to form an enhanced interfacial interaction between PP and SCF, meanwhile the crossover sites of adjacent SCFs were locked by the cured EP component to form a strong network.

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1. Introduction

With the demand of high-performance polymeric materials, short fiber reinforced polymer composites (SFRP) have attracted increasing interest for researchers in the field of materials science from both academia and industry [1–3]. According to the source and chemical composition, the short fiber can be classified into glass fiber, carbon fiber, aramid fiber, Kevlar fiber and natural fiber. Among all of these fibers, short carbon fiber (SCF) was more attractive due to its advantages with light mass, high conductivity, and superior reinforce-effect for polymer matrices [4,5].

Polypropylene (PP) is one of the most widely used common plastics with widely applications in the field of packaging, textiles, automobiles, electric industry, etc. [6–8]. However, its application as engineering plastics is greatly restricted due to its low modulus and strength. Therefore, considering the advantages of SCF, a feasible way to improve the stiffness of PP is to prepare PP/SCF composites. Hitherto much excellent research work has been done in PP modification and novel material exploitation by incorporating SCF. For example, Fu et al. [9] investigated the effects of fiber content and length on tensile properties of PP/SCF composites. Rezaei et al. compared mechanical properties of PP/SCF composites with steel to choose for car bonnet application [10,11].

It is well known that the interfacial adhesion between fillers and polymer matrix was one of the key factors to determine the structure and properties of obtained polymer composites [12,13]. Maleic anhydride grafted polypropylene (PPMA) is generally employed as compatibilizer to improve the interfacial adhesion between PP and filler [14–16]. Karsli et al. used PPMA to enhance the interfacial adhesion between SCF and PP, and the ultimate tensile strength and modulus of modified PP composites were higher than those of SCF directly reinforced PP composites [17]. This may be ascribed to the reaction between PPMA and epoxy resin (EP) on the surface of SCF. As we know, EP is often used as an important component of sizing agent to improve surface properties of carbon fiber [18,19]. However, the incompatibility between PP and epoxy resin will damage the performance of the composite. Recently Ha et al. [20] have confirmed the generation of an ester linkage from the reaction between the anhydride group of anhydride grafted polyethylene and the hydroxyl group of epoxy resin by FTIR spectra. Jiang et al. [21] found that the addition of PPMA improved interfacial adhesion between PP and EP, therefore impact strength and tensile strength of PP/EP blends were increased.

Although the addition of PPMA can promote the mechanical properties of PP/SCF composites, the improvements are still not enough to meet the demand as engineering plastics. This might result from the characteristic of weak interaction between crossover sites of different SCFs in the matrix. As a result, the effective network structure of SCF for bearing load was not formed in the

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PP composites. We wonder whether the formation of network structure of SCF in PP matrix can further promote the mechanical properties of PP/SCF composites by adding PPMA and EP together. Here PPMA can improve the interfacial interaction between PP and SCF, and the added EP is expected to lock crossover sites of different SCFs in the matrix. In this work, EP was introduced into PP/PPMA/SCF system, and an *in-situ* cured technology of EP for locking crossover sites of different SCFs was developed during the melt compounding. Thus a strong network structure of SCF was formed in PP matrix. As a result, the mechanical properties of the resultant composites were much higher than those of PP/PPMA/SCF composites. The improved mechanical properties of PP/PPMA/EP/SCF composites were ascribed to the synergistic effect between PPMA and EP in the composites. The synergistic reinforcing mechanism was also discussed on the base of SEM observation, FTIR analysis, extraction test and rheological measurement. According to our best knowledge, there is no report in which both PPMA and EP were added together to modify PP/SCF composites.

2. Experimental sections

2.1. Materials

Polypropylene powder (PP, isotactic, T30s, $M_w = 2.89 \times 10^5$ g/mol⁻¹, PDI = 3.45, melt flow index at 230 °C: 6.8 g/min) was purchased from Panjin Petrochemical Co. The maleic anhydride grafted polypropylene (PPMA, $M_n = 7.43 \times 10^4$ g/mol⁻¹, PDI = 2.73, melt flow index at 230 °C: 32 g/min) with 0.8 wt% of MA group was supplied by Starbetter (Beijing) Chemical Materials Co. Short carbon fiber (SCF, 3 mm in average length, epoxy resin sizing agent of 2.8 wt%) was purchased from Shanghai Yingjia special fiber material Co.

Epoxy resin (EP, diglycidyl ether of bisphenol-A) with epoxy value of 0.51 was supplied by Bluestar New Chemical Materials Co. The curing agent imidazole was reagent-grade obtained from Tianjin Chemical Co. All the materials except for imidazole were dried in a vacuum oven at 80 °C for 12 h before using.

2.2. The preparation of PP composites

The composites were prepared in a Brabender mixer (Banbury mixer) by melt compounding. The temperature and the rotor speed were set at 180 °C and 50 rpm, respectively. In a typical experiment, PP or PP/PPMA was melted, and then EP at 4 min, SCF at 6 min and imidazole at 8 min was in turn added to the melt, respectively. The mixture was removed at 12 min and then cut into small pieces and cooled. The dosage of imidazole (as curing agent for EP) was 4 wt% of EP [21,22]. Table 1 shows the formulation of PP/SCF composites with various compositions.

The mixture was first dwelled at 190 °C for 2 min in a hot press, and then compressed at 15 MPa for 2 min. Two picture frames (1 mm and 4 mm thickness, with 15×15 cm² hole) were used to constrain the composite deformation. After that the frame was transferred to a cold press and compressed at 10 MPa until the sample reached room temperature. Finally composite sheets (15×15 cm²) with 1 and 4 mm thickness were fabricated,

respectively. Then the sheets were cut by mould into the provision shape for mechanical property testing. For FTIR characterization, the PP, PPMA and PP composites were pressed into film with a thickness less than 20 µm, and EP was directly tested by dropping the liquid onto a KBr sheet.

2.3. Characterization

Uniaxial tensile tests were performed with an Instron 1121 testing machine (Canton, MA) according to ISO 527-2 standard. Specimens ($10 \times 2 \times 1$ mm³) were cut from the previously compression-molded sheet into a dumbbell shape. The measurements were conducted at a crosshead speed of 1 mm/min at room temperature. The flexural properties were tested under three-point bending in a DXLL-5000 machine (Shanghai Jiedeng Instruments Ltd., China) in accordance with ISO 178:2003. The size of the flexural testing samples used was $80 \times 10 \times 4$ mm³. The machine was operated at a crosshead speed of 1 mm/min and a span length of 60 mm. The V-notched specimens ($80 \times 8 \times 4$ mm³) were tested to measure the impact strength according to ISO 180 procedures with an impact testing machine (CEAST, Chengde, China). All tests were carried out at room temperature and 50% relative humidity. At least five runs for each sample were measured, and the results were averaged.

The tensile-test specimens fracture surface and residues of composite materials after solvent extraction with xylene were observed under XL30ESEM FEG field emission scanning electron microscope to study interfacial adhesion.

FTIR (Bio-Rad FTS-135) was used to characterize composite samples, which were pressed into thin films at 185 °C. Rheological measurements were conducted in a Physcia MCR300 rotation rheometer under nitrogen atmosphere. Round samples (25 mm in diameter $\times$ 1 mm in thickness) were prepared for frequency scanning at 185 °C, with a gap of 0.8 mm and a frequency scope from 0.05 to 100 rad/s, and a strain of 1% was used, which was in the linear viscoelastic regime for all samples.

The composite was placed in a muffle furnace at 500 °C for 60 min. The matrix resin was decomposed under this temperature and the carbon fibers were remained. The length distributions of remaining carbon fibers were quantified using an image analysis technique. The fibers were dispersed in ethanol and then coated on the glass slide. Images of various fiber lengths were taken by a Panasonic WV-CP280 Closed-circuit surveillance camera equipped on Shanghai Jinke XTZ-E continuous variable times microscope. The images were then analyzed using Nano Measurer processing software. A series of 1000 samples were measured for remaining carbon fibers. The distribution and polydispersity of the fiber length were then calculated through the formula below:

$$\text{Number average length, } L_N = \frac{\sum N_i L_i}{\sum L_i} \quad (1)$$

$$\text{Weight average length, } L_W = \frac{\sum N_i L_i^2}{\sum N_i L_i} \quad (2)$$

$$\text{Polydispersity, } d = \frac{L_W}{L_N} \quad (3)$$

Table 1

The formulation of PP/SCF composites with various composition.

Sample	PP (wt%)	PPMA (wt%)	EP (wt%)	SCF (wt%)
PP	100	0	0	0
PP/SCF	73.6	0	0	26.4
PP/PPMA/SCF	66.1	7.5	0	26.4
PP/PPMA/EP/SCF	54.5	7.5	11.6	26.4
PP/EP/SCF	62.0	0	11.6	26.4

The dosage of imidazole (as curing agent for EP) was 4 wt% of EP.

3. Results and discussion

3.1. Effects of EP and PPMA on mechanical properties of PP/SCF composites

The tensile properties of neat PP and PP/SCF composites were initially investigated. As shown in Fig. 1a, it is apparent that the

tensile strength and Young's modulus of PP/SCF binary composites are slightly higher than those of pure PP due to the reinforcing effect of added SCF. When PPMA was added into PP/SCF to improve the interfacial interaction between PP and SCF, the tensile properties were increased to some extent on the base of PP/SCF. However, when EP alone was added into PP/SCF, the tensile strength of PP/EP/SCF composites decreased and the Young's modulus increased compared to those of pure PP and PP/SCF composites. Interestingly, when both PPMA and EP were added into PP/SCF system, the resultant composites (PP/PPMA/EP/SCF) exhibited a significant increase on the tensile strength and Young's modulus, which are 273% and 239% in comparison with those of neat PP. These results suggested a synergetic effect between PPMA and EP on improving the tensile strength and Young's modulus of PP/SCF composites. However, compared to pure PP (140%), tensile strain at break of all PP composites dramatically decreased (<5%), which was attributed to the presence of SCF in PP matrix [23].

Further investigations confirmed that the combination of PPMA and EP also exhibited a synergetic effect on improving the flexural properties of PP/SCF composites. As shown in Fig. 1b, the incorporation of SCF alone, PPMA/SCF, and EP/SCF into PP showed no significant reinforced effect on flexural strength, although they were effective to improve the flexural modulus of PP, especially in the case of incorporating EP/SCF. However, compared to that of neat PP, the incorporation of PPMA/EP/SCF could dramatically improve both the flexural strength and modulus, in which the increments achieved to 351% and 624%, respectively.

The toughness of obtained PP composites was estimated by notched impact test, which is also an important parameter for mechanical properties of polymer materials. Although the addition of SCF, PPMA/SCF, and EP/SCF into PP could slightly improve the impact strength, the PP/PPMA/EP/SCF displayed the highest impact strength, which is over 336% of that for neat PP (Fig. 1c). This further confirmed the synergistic effect of the combination between

PPMA and EP. It should be mentioned that imidazole was added as curing agent for EP in the system containing EP, so EP was cured (or partially cured) in the resultant composites. As a control, when imidazole was not added as curing agent for EP in the system, the resultant composites showed very poor mechanical properties due to EP existing as small molecules in the system.

3.2. Effect of PPMA and EP on interfacial adhesion of PP/SCF

In order to investigate the relationship between the structure and performance of PP/SCF composites, SEM was employed to elucidate the microstructure and morphology of PP/SCF composites (Fig. 2). In the PP/SCF composite, the SCF filaments were uniformly distributed in the PP matrix (Fig. 2a), but the surface of pulled-out SCF was smooth and some cavities were clearly visible due to poor interfacial adhesion [17] (Fig. 2b). In the PP/PPMA/SCF system, the cavities were still visible, but the surfaces of pulled-out SCF filaments became rough (Fig. 2c and d), indicating that there is some interaction between PPMA and SCF, which resulted from possible chemical reaction [24]. In the PP/EP/SCF composites, the SCF filaments present a slightly rough surface (Fig. 2e and f), which might result from the absorption of EP. However, there was clear void between some SCF filaments and PP matrix (see the area marked by dashed line), which indicated poor interfacial adhesion between SCF and PP matrix. In contrast, when both PPMA and EP were added into PP/SCF, the SCF filaments became short and fuzzy, and few cavities were observed (Fig. 2g). On the amplified image (Fig. 2h), it is apparent that the SCF filaments were completely encompassed by polymers. These results implied a strong interfacial adhesion between SCF and PP matrix in this case [25].

To further evaluate the interfacial adhesion between SCF and PP matrix, the PP/SCF composites with different formulations were extracted by xylene, and the remaining solids were observed by SEM (Fig. 3). As shown in Fig. 3a from PP/SCF, the surfaces of SCF

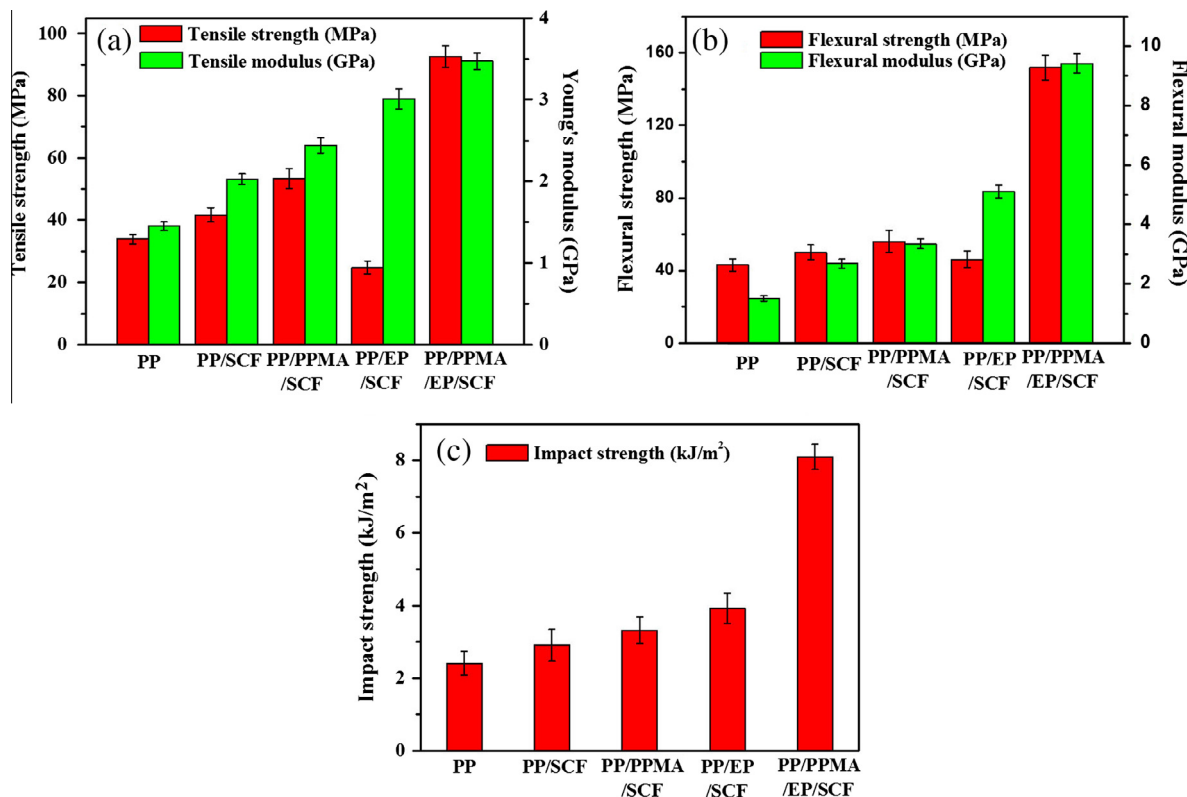


Fig. 1. The mechanical properties of different components composites: (a) tensile properties; (b) flexural properties; and (c) impact strength.

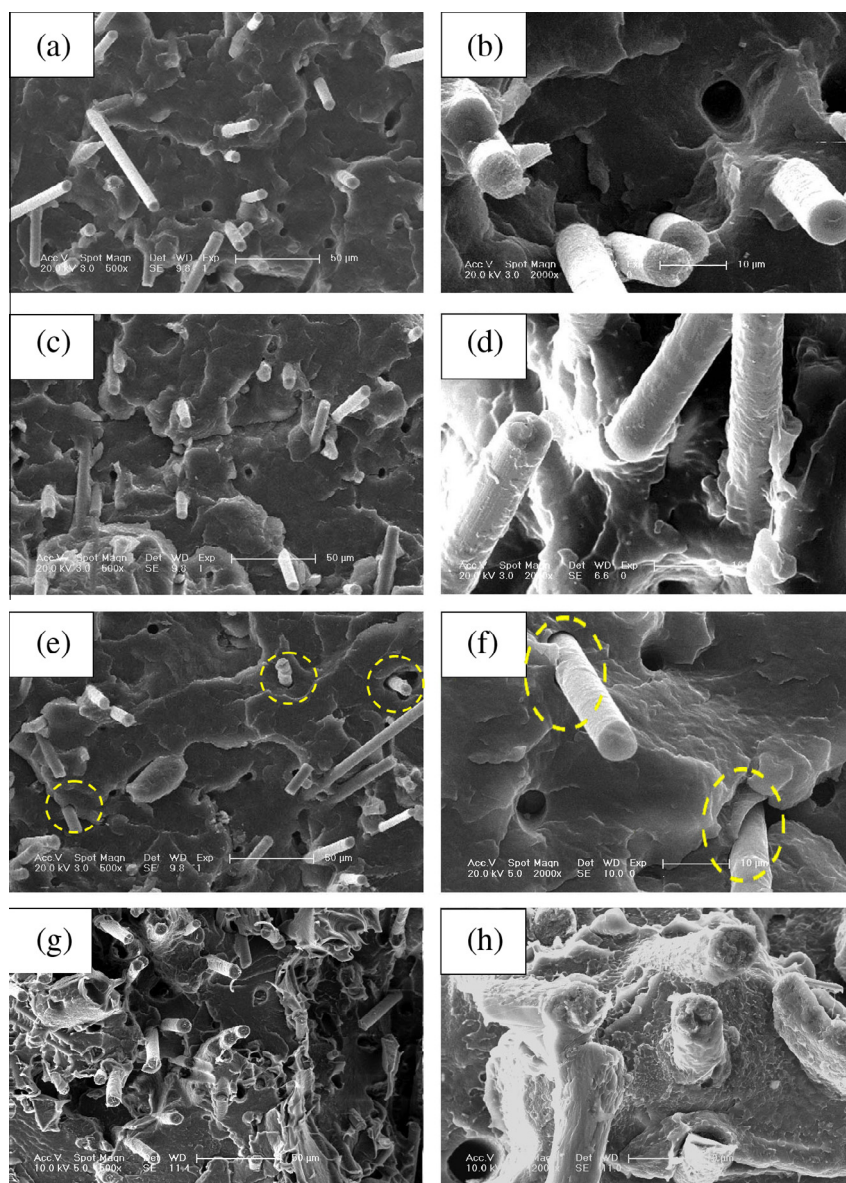


Fig. 2. SEM micrographs of fractured surfaces of tensile specimen: (a) and (b) PP/SCF; (c) and (d) PP/PPMA/SCF; (e) and (f) PP/EP/SCF; and (g) and (h) PP/PPMA/EP/SCF.

were smooth and only few small particles were left, indicating a weak adhesion between PP and SCF. For PP/PPMA/SCF sample, some spherical polymer residues were left on the surfaces of SCF (Fig. 3b), which may result from the residual PPMA due to esterification reaction between the hydroxyl groups on the surfaces of SCF and PPMA [24,26]. Similarly, for PP/EP/SCF, some block-like residues were left on the SCF surface (Fig. 3c), which may be ascribed to the cured EP component. However, the residues from PP/PPMA/EP/SCF were totally different. The SCF surfaces were almost completely covered by large amounts of solid residues (Fig. 3d). This result indicated that there was the strongest interfacial adhesion between polymer matrix and SCF in the PP/PPMA/EP/SCF composite, which was responsible for the best mechanical properties of PP/PPMA/EP/SCF composite among all PP/SCF composites.

3.3. Synergistic mechanism between PPMA and EP in PP/SCF composites

According to the above results, the enhanced mechanical properties of PP/PPMA/EP/SCF were mainly attributed to the synergistic

effect between PPMA and EP. However, it is a question what the synergistic mechanism between PPMA and EP on improving the mechanical properties is. A possible reason is that the presence of both PPMA and EP could effectively improve the interfacial adhesion between PP and SCF. Taking into account the chemical structures of PPMA and EP, the anhydride from PPMA may react with the epoxy group [27,28]. To confirm this possible reaction, FTIR measurements for PPMA, EP and PPMA/EP melt mixture (PPMA-EP) were carried out, respectively. As shown in Fig. 4, the peaks at 913 and 761 cm^{-1} in the FTIR spectra of EP were assigned to the characteristic absorptions of epoxy group [29,30], and the peaks at 1782 cm^{-1} in the FTIR spectra of PPMA was corresponded to the stretching vibration of anhydride [31]. However, the intensities of these peaks in the FTIR spectra of PPMA/EP mixture were greatly decreased. Meanwhile, a strong new peak at 1728 cm^{-1} appeared which was assigned to the stretching vibration of carbonyl group in ester compounds [32]. These results proved that epoxy group from EP reacted with the anhydride from PPMA during melt compounding. Fig. 5 showed the chemical structure of EP and possible chemical reactions between EP and PPMA. Firstly the

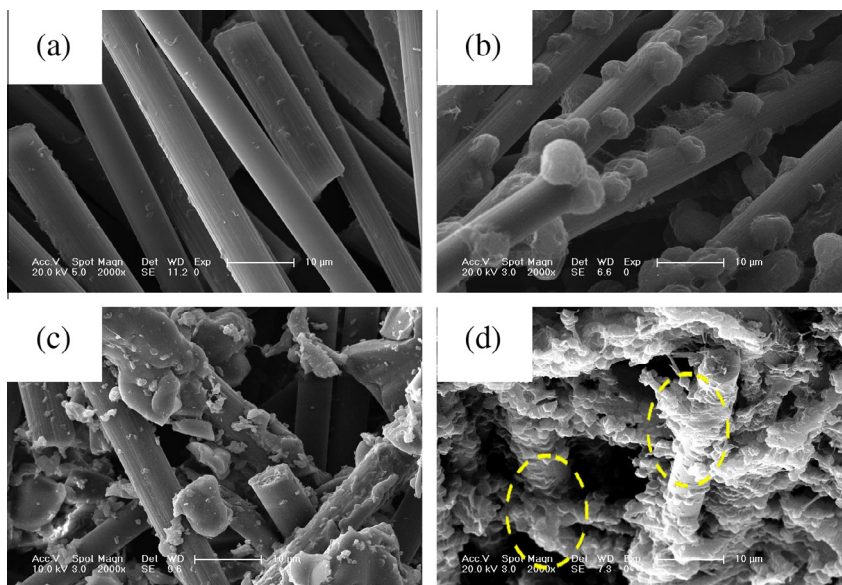


Fig. 3. SEM micrographs of residues on the surfaces of SCF after xylene extraction: (a) PP/SCF; (b) PP/PPMA/SCF; (c) PP/EP/SCF; and (d) PP/PPMA/EP/SCF.

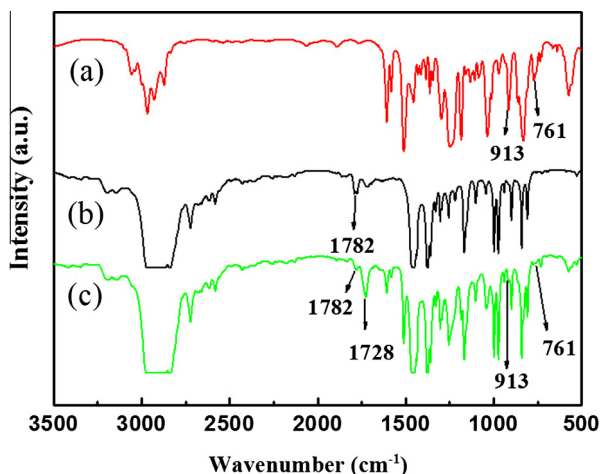


Fig. 4. FTIR spectra of EP (a), PPMA (b) and PPMA/EP melt mixture (c).

hydroxyl groups of EP could react with the anhydride group from PPMA to form ester bond and new carboxyl group at the same time. Then this carboxyl group could react with epoxy group to form another ester bond, and release a new hydroxyl group, which continue to take part in the reaction with the anhydride. As a result, a large amount of PPMA was grafted onto the EP component.

To further investigate the synergistic mechanism between PPMA and EP, the rheological measurements of neat PP and PP composites were also performed. As shown in Fig. 6, when only SCF was added into PP, the complex viscosity and storage modulus were increased, especially in the low frequency. This means that the SCF formed network-like structure in the PP matrix [33,34]. When both PPMA and SCF were added into PP matrix, anhydride group of PPMA reacted with hydroxyl group on the surface of SCF (from the sizing of SCF). Thus PPMA acted as a compatibilizer in PP/SCF composites. As a result, the interfacial adhesion between SCF and PP matrix was strengthened, which will result in relatively higher complex viscosity and storage modulus than those of PP/SCF

composites. On the other hand, the addition of PPMA with low molecular weight may result in the decrease of the complex viscosity and storage modulus of the PP melt. Owing to counteracting effect, compared to PP/SCF composites, the complex viscosity and storage modulus of the PP/PPMA/SCF composites changed slightly.

When EP was added into PP/SCF, as PP is not miscible with EP, a part of EP component prefers to cover the surface of SCF, due to the sizing of SCF surface containing epoxy resin component, and the others will form isolated phase in PP matrix. The remaining solids after the extraction for PP/EP/SCF showed that the aggregates of cured EP were found at the crossover sites of adjacent carbon fibers (Fig. 3c). Several adjacent carbon fibers were connected with the aggregates of cured EP, leading to the formation of network structure. Therefore the melt viscosity and storage modulus of this sample were significantly increased (compared to PP/SCF composites, Fig. 6). However, in the case of PP/EP/SCF composites, although adjacent carbon fibers could form a good network structure, owing to the weak interfacial interaction between PP matrix and SCF (or EP), stress transfer efficiency was low during mechanical tests. Thus the tensile strength was also low comparing with those of PP/SCF.

When PPMA and EP were added into PP/SCF system at the same time, on one hand, a layer of EP was covered on the surface of SCF, which increased the active sites on carbon fiber surface and helped SCF to form network structure. On the other hand, the presence of PPMA improved the interfacial interaction between PP matrix and EP component, which could promote stress transfer efficiency from PP substrate to SCF (or EP layer on the surface of SCF). So the viscosity and storage modulus of PP/PPMA/EP/SCF increased, compared to the PP/EP/SCF composites. The extraction experiments of PP/PPMA/EP/SCF composites showed that the crossover sites of SCF were locked by crosslinked EP component (Fig. 3d), which promoted the formation of SCF network structure. The formation of SCF network structure, in which the crossover sites are strongly locked, will dramatically increase the mechanical properties of the composites (Fig. 1). Rowan and coworkers have found that strong interaction between percolating rigid nanofibers can maximize stress transfer and therewith the overall modulus of the nanocomposites [35]. In this work, compared to other samples, the formation of SCF network structure with locked crossover sites